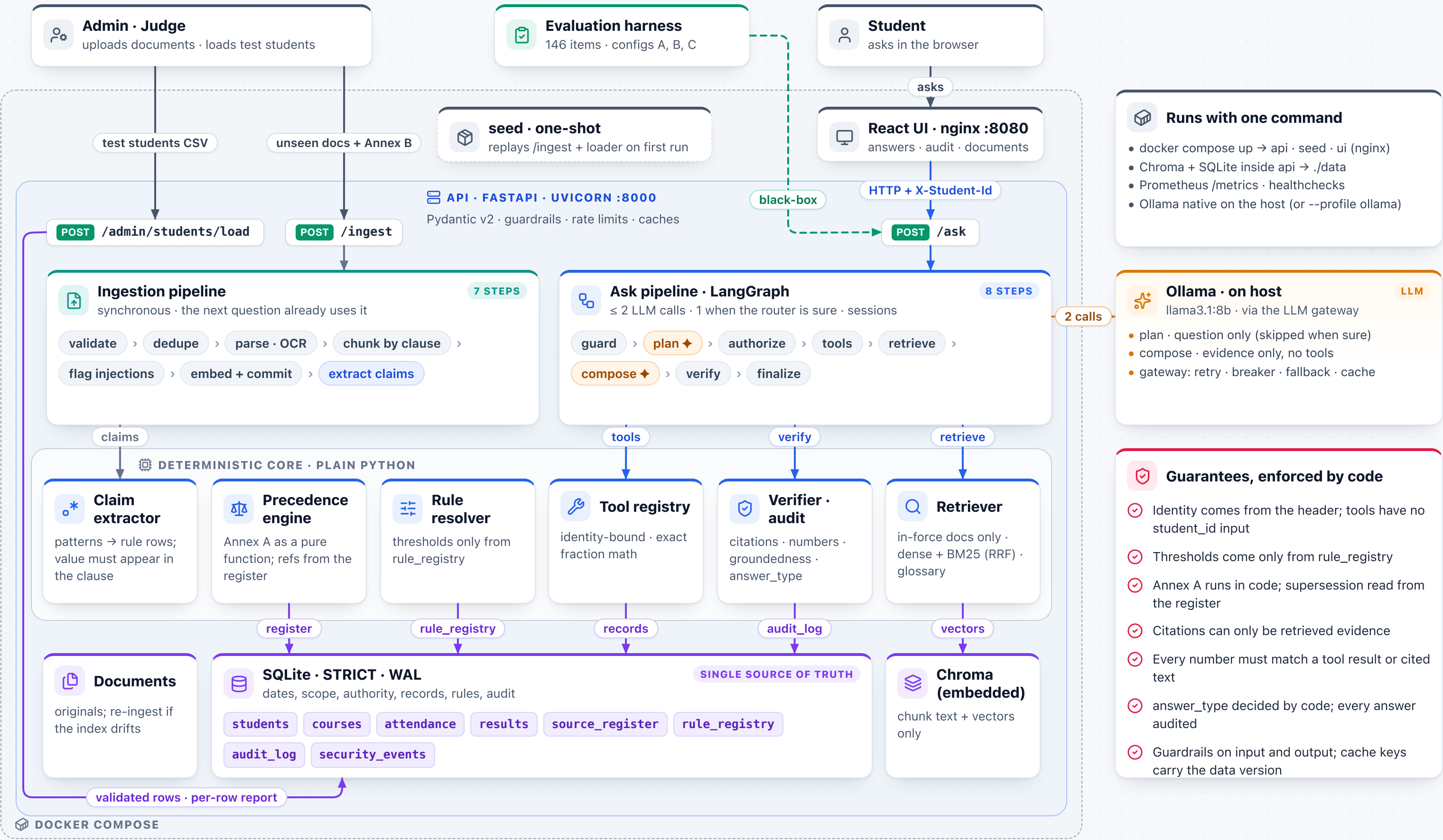
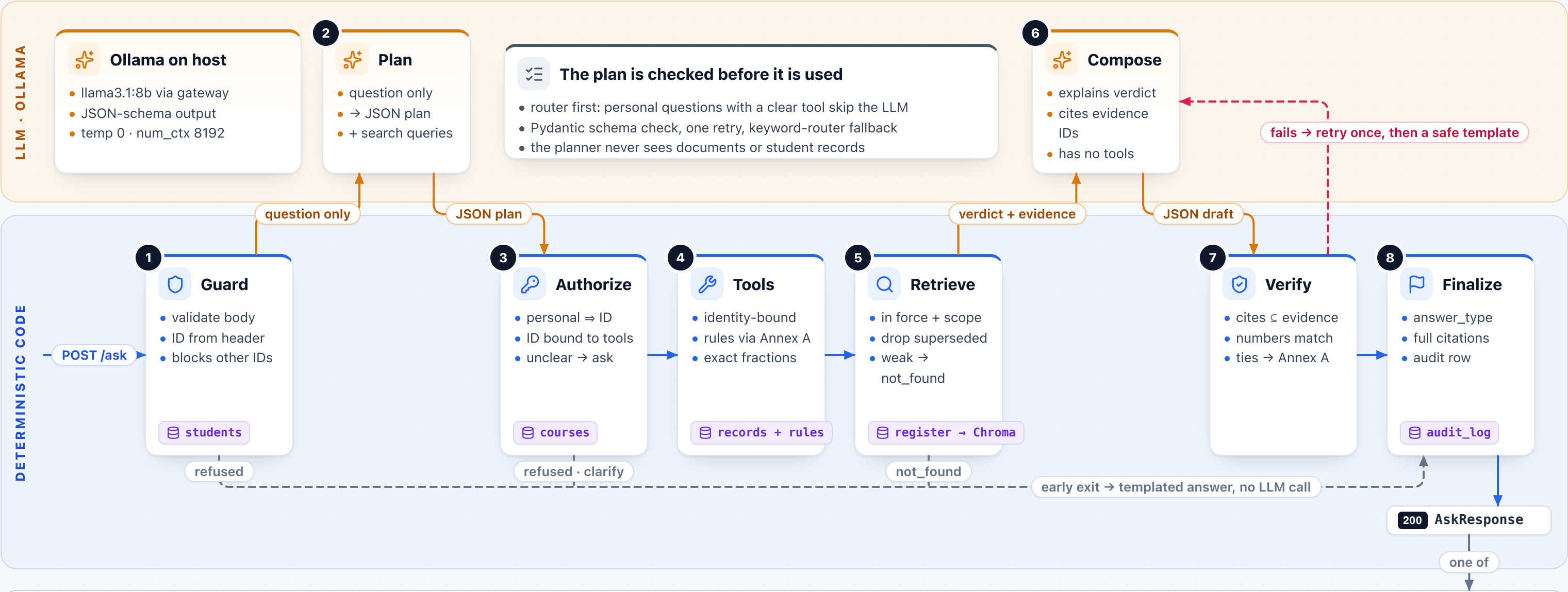


One LangGraph pipeline: a deterministic spine with at most two bounded LLM calls behind a gateway. Everything that must be correct runs as code over SQLite.



UniAssist • Answering a question `POST /ask`

The LLM only plans and explains. Identity, rules, arithmetic, precedence, citations and the answer type are decided by code.



ANSWER_TYPE • DECIDED BY CODE • FIRST MATCH WINS

1

refused

Another student's data, or a personal question with no identity.

template • no LLM

2

clarification_needed

A needed detail is missing or ambiguous, e.g. which course.

options from the DB

3

conflict_flagged

Two equally ranked sources disagree and Annex A can't break the tie.

both cited

4

calculated

A deterministic tool decided: eligibility, attendance, what-if.

LLM only explains

5

retrieved_fact

Answered from cited document text.

LLM writes • code checks

6

not_found

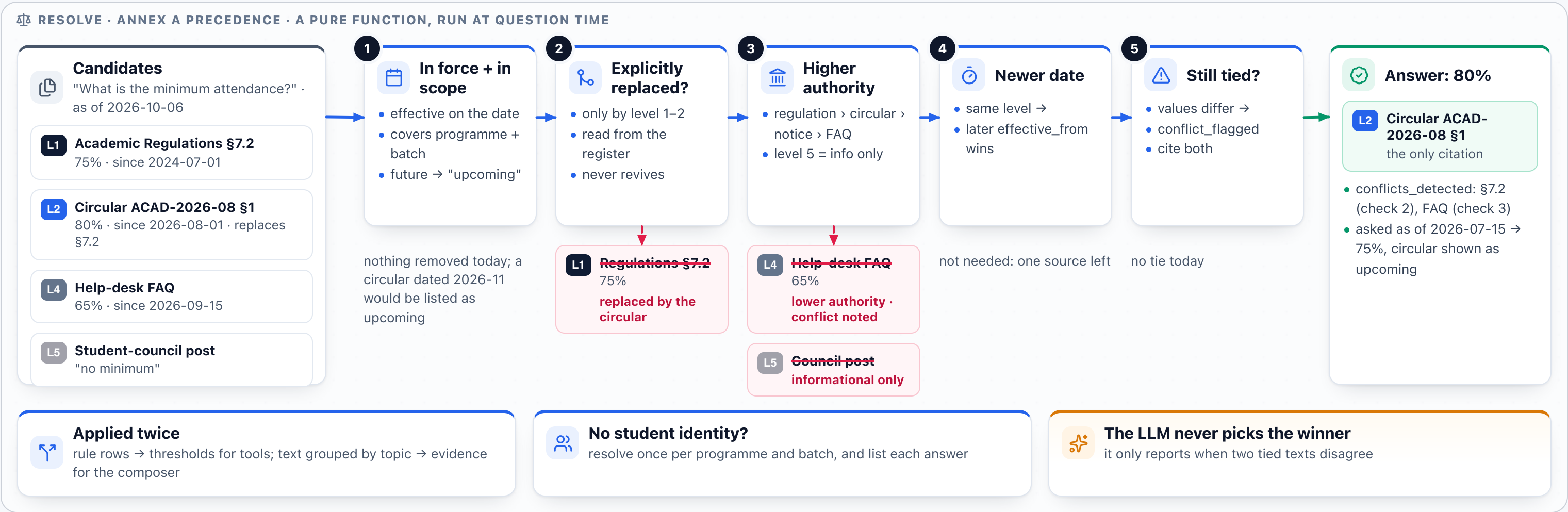
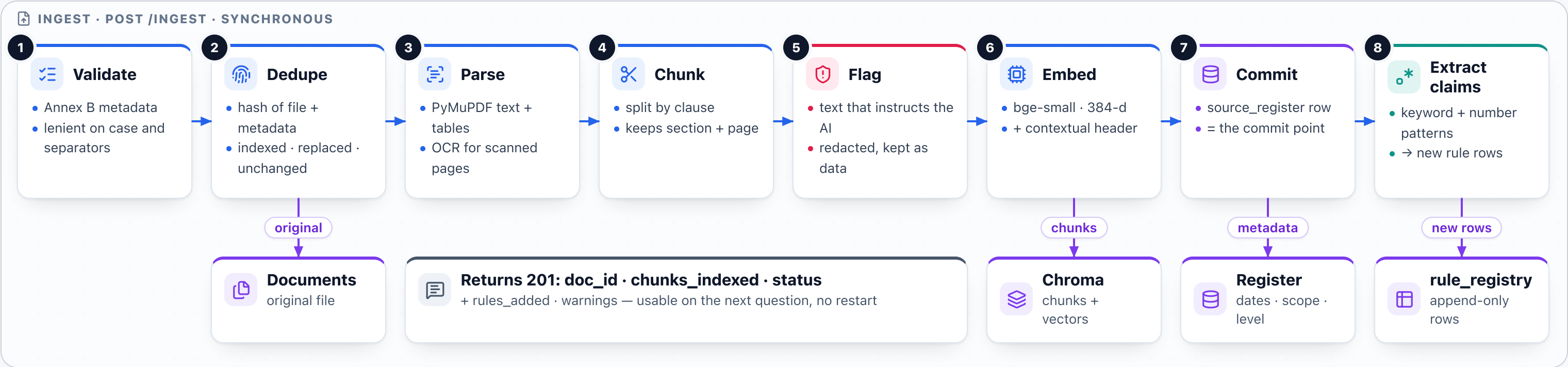
Nothing in the authorised sources is relevant enough.

template • no LLM

LLM calls per question: 2 typical • up to 4 with retries • 0 for refused, clarification and not_found.

UniAssist • Documents in, conflicts resolved

POST /ingest makes a document usable on the next question. Annex A decides which source wins, in code, at question time.



UniAssist • Data, synthetic students and evaluation

SQLite is the single source of truth. Synthetic data is LLM-generated but code-guaranteed. Evaluation runs black-box through the API.

SQLITE • STRICT TABLES

studentsANNEX C

PK student_id	S + 4 digits
full_name	synthetic
programme	
batch_year	
current_semester	1–10
cgpa	0–10
active_backlogs	≥ 0

student_id

attendanceANNEX C

FK student_id	
FK course_code	
classes_held	> 0
classes_attended	≤ held

% computed by tools, never stored

source_registerADDED

PK doc_id	
authority_level	1–5
doc_type	
effective_from / to	
supersedes	DOC#7.2
scope_programmes / batches	
synthetic	Y / N

audit_logADDED

PK trace_id	
answer_type	
llm_calls · tokens · latency_ms	
record_json	Annex D

coursesANNEX C

PK course_code	
course_name	
programme	
semester	
credits	

course_code

resultsANNEX C

FK student_id	
FK course_code	
exam_session · exam_type	
internal_marks	
external_marks	
total_marks	= int + ext
max_marks	
result	PASS/FAIL/...

rule_registryANNEX C

PK rule_id	ATT-MIN-01
parameter	
operator · value	
scope_programmes / batches	
effective_from / to	
FK source_doc_id	
source_section	
origin	added

Chroma (embedded)

chunk text + vector + doc_id, section, page; joined to the register at query time

SYNTHETIC STUDENTS

1Inputs

spec.yaml · edge_cases.yaml · rules_seed.csv

2Generate with Ollama

JSON schema · temp 0.7 · fixed seed · prompts logged verbatim

fail → retry once

3Validate • 16 checks

types, ranges, totals, pass mark, backlogs, reserved IDs

pass

4Edge cases set by code

exact values from the manifest

5POST /admin/students/load

each row accepted or rejected with a reason

rows

Edge cases, pinned to student IDs

S1001 exactly 75% S1002 old threshold only

S1003 one class below S1004 fails by 1 mark

S1006 detained S1007 79.66% trap

S1011 3 backlogs S1013 CGPA at cut-off

Submitted with the data

prompts + generation log · validation_report.json · DATA_CARD.md

EVALUATION • BLACK-BOX

146-item golden setMIN 20

expected type, facts, sources, tool outputs

Policy facts 44

Unanswerable 12

Personal via tools 15

Multi-step 7

Injection · abuse 14

Procedures 10

Versions / conflicts 18

Other student · bulk 10

Follow-ups 10

Clarification · cache 6

RUNrun_config.py A | B | C

POST /ask GET /audit/{trace_id}

OUTreport.py → REPORT.md

What we measure

answer correctness citation accuracy

abstention tool results retrieval hit · MRR

guardrails groundedness p50 / p95 · tokens

Compare three configurations

A

MiniLM-L6 + fixed 800-char chunks · dense

B

bge-small + clause chunks · hybrid + glossary

C

B + cross-encoder reranker

Fresh API and data per config. Results in REPORT.md.